

How to Scale Your Model

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All About Rooflines:

Question 1.1

We need to load 1 byte per value x_{ij} , so for matrix $X[B,D]$ we need to load BD bytes and for $Y[D,F]$ we need to load DF bytes. Then, once the matrix is finished, we need to write BF bytes back into memory as $Z[B,F]$.

Thus, total byte transfers is

$$BD + DF + BF.$$

Question 1.2

We are performing one OP per x_{ij} and y_{ij} multiplication, and thus BDF multiplications. We also perform $BF(D-1)$ additions. Therefore, we perform $BDF + BF(D-1) = BF(2D-1)$ total OPs

Example: $X[2,3]$, $Y[3,3]$.

Let

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{bmatrix}, \quad Y = \begin{bmatrix} y_{11} & y_{12} & y_{13} \\ y_{21} & y_{22} & y_{23} \\ y_{31} & y_{32} & y_{33} \end{bmatrix}.$$

Then

$$Z = XY = \begin{bmatrix} x_{11}y_{11} + x_{12}y_{21} + x_{13}y_{31} & x_{11}y_{12} + x_{12}y_{22} + x_{13}y_{32} & x_{11}y_{13} + x_{12}y_{23} + x_{13}y_{33} \\ x_{21}y_{11} + x_{22}y_{21} + x_{23}y_{31} & x_{21}y_{12} + x_{22}y_{22} + x_{23}y_{32} & x_{21}y_{13} + x_{22}y_{23} + x_{23}y_{33} \end{bmatrix}.$$

Question 1.3

Arithmetic intensity:

$$\text{Arithmetic Intensity} = \frac{\text{Computation OPs}}{\text{Communication bytes}} = \frac{BF(2D-1)}{BD + DF + BF}.$$

Assuming batch size $B \ll D, F$,

$$\frac{BF(2D-1)}{BD + DF + BF} \approx \frac{BF(2D)}{DF} = 2B.$$

As mentioned in the matrix-multiplication section, equation (9):

$$\text{Intensity}(\text{Matmul}) > \text{Intensity}(\text{TPU}).$$

$$\therefore 2B > \frac{3.94 \times 10^{14}}{8.2 \times 10^{11}} \approx 480.$$

Thus

$$2B \gtrsim 480 \quad \Rightarrow \quad B \gtrsim 240.$$

Therefore, when doing int8 matmul, for batch size $B \gtrsim 240$ tokens we start to become compute-bound.

Question 1.4

Define

$$T_{\text{lower}} = \max(T_{\text{math}}, T_{\text{comms}}), \quad T_{\text{upper}} = T_{\text{math}} + T_{\text{comms}}.$$

Compute times:

$$\begin{aligned} T_{\text{math}} &= \frac{\text{Computation OPs}}{\text{Accelerator OPS/s}} = \frac{BF(2D-1)}{3.94 \times 10^{14}}, \\ T_{\text{comms}} &= \frac{\text{Communication bytes}}{\text{Bandwidth bytes/s}} = \frac{BD + DF + BF}{8.2 \times 10^{11}}. \end{aligned}$$

Therefore

$$\begin{aligned} T_{\text{lower}} &= \max\left(\frac{BF(2D-1)}{3.94 \times 10^{14}}, \frac{BD + DF + BF}{8.2 \times 10^{11}}\right), \\ T_{\text{upper}} &= \frac{BF(2D-1)}{3.94 \times 10^{14}} + \frac{BD + DF + BF}{8.2 \times 10^{11}}. \end{aligned}$$

To get a more specific answer for T_{lower} , we can cross-multiply terms.

If $T_{\text{math}} > T_{\text{comms}}$ then

$$\frac{\text{Computation OPS}}{\text{Communication Bytes}} > \frac{\text{Accelerator OPS/s}}{\text{Bandwidth Bytes}}.$$

which means

$$\frac{BF(2D-1)}{BD + DF + DF} > \frac{3.94 \times 10^{14}}{8.2 \times 10^{11}}.$$

$$\approx \frac{2BDF}{DF} \approx 2B > 480,$$

which implies

$$B > 240.$$

Therefore:

- When $B > 240$, $T_{\text{lower}} = T_{\text{math}}$.
- When $B < 240$, $T_{\text{lower}} = T_{\text{comms}}$.

Question 2

This is essentially the same scenario, except the communication bytes loaded changes

In the scenario in question 1, we loaded 1 byte per matrix value in $X[B,D]$ and $Y[D,F]$, Now, we're loading 2 bytes per value in $\text{bf16}[B,D]$ and 1 byte per value in $\text{int8}[D,F]$

$$\text{Communication bytes} = 2 \cdot BD + 1 \cdot DF + 2 \cdot BF$$

where $2 \cdot BD + 1 \cdot DF$ are load bytes and $2BF$ are write bytes

$$\text{FLOPs/s} = 1.97e14$$

$$\text{FLOPs} = BF(2D - 1) \approx 2BDF \text{ *nothing changes from previous scenario}$$

$$\text{Intensity} = \frac{2BDF}{2BD + DF + 2BF} \approx \frac{2BDF}{DF}$$

*assuming $2B \ll D$ and $2B \ll F$

$$= 2B$$

$$\text{Intensity (TPU)} = \frac{1.97e14}{8.2e11} = 240$$

With the assumption that $2B \ll D, F$, we get a similar result to the $\text{bf16}[B, D] \times \text{bf16}[D, F]$ matmul result: we become compute bound around $2B > 240$ and $B > 120$

Question 3

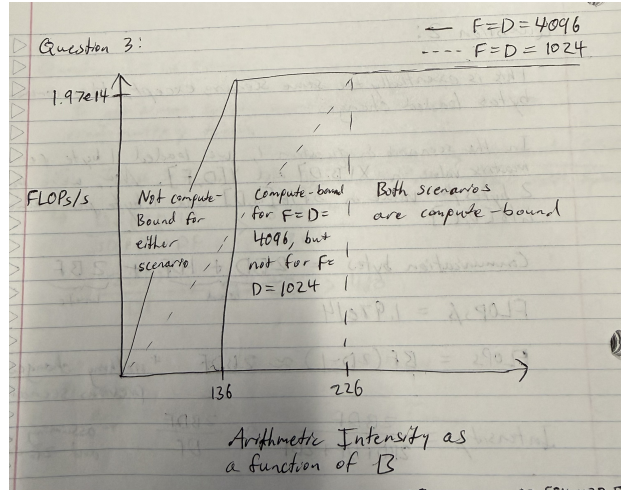


Figure 1: Question 3 – Boundedness for different batch sizes

For $F=D=4096$:

$$\begin{aligned} \text{Intensity}(\text{matmul}, F = D = 4096) &= \frac{2BDF}{2BD + DF + 2BF} \\ &= \frac{33,554,432 B}{16,384 B + 16,777,216} \end{aligned}$$

Border at Intensity (matmul, F= D=4096) = Intensity (TPU)

$$\frac{33,554,432 \text{ B}}{16,384 \text{ B} + 16,777,216} = 240$$

$$33,554,432 \text{ B} \approx (4e6) \text{ B} + 4e9$$

$$3e7 \text{ B} = 4e9$$

$$\text{B} \approx 136$$

For F=D=1024:

$$\begin{aligned} \text{Intensity}(\text{matmul}, F = D = 1024) &= \frac{2\text{BDF}}{2\text{BD} + \text{DF} + 2\text{BF}} \\ &= \frac{2,097,152 \text{ B}}{4096 \text{ B} + 1,048,576} \end{aligned}$$

Border at Intensity (matmul, F=D=1024) = Intensity (TPU)

$$\begin{aligned} \frac{2,097,152 \text{ B}}{4096 \text{ B} + 1,048,576} &= 240 \\ 2,097,152 \text{ B} &= 983,040 \text{ B} + 251,658,240 \\ (1e6) \text{ B} &= 251,658,240 \\ \text{B} &\approx 226 \end{aligned}$$

Question 4

$$\text{Arithmetic Intensity} = \frac{\text{FLOPs}}{\text{Comms Bytes}}$$

$$\text{Communication Bytes} = \text{BD} + \text{BDF} + \text{BF}$$

If the matrix is different for each batch element, that doesn't change the number of FLOPs since we still have the same number of unique products $x_{ij} y_{ij}$, except y_{ij} just changes each batch element.

$\therefore$ FLOPs is the same as in question 1:

$$\begin{aligned} &\text{BF}(2D - 1) \\ \text{Intensity} &= \frac{\text{BF}(2D - 1)}{\text{BD} + \text{BDF} + \text{BF}} \end{aligned}$$

Question 5

According to the spec sheet,

$$\text{BFLOAT16 Tensor Core} = 1979 \text{ teraFLOPs/s}$$

divide by 2 to get true value (only achievable with structured sparsity)

$$\text{bfloat16} \Rightarrow \frac{1979}{2} \text{ teraFLOPs/s}$$

$$\text{GPU Memory Bandwidth} = 3.35 \text{ TB/s}$$

bfloat16 matmul becomes compute-bound when

$$\text{Intensity (matmul)} > \frac{1979}{\frac{3.35}{2}} \approx \frac{591}{2} = 295$$

we calculated bfloat matmul intensity in the matrix multiplication section to be

$$\approx B$$

and more accurately to be $= \frac{\text{BDF}}{\text{BD}+\text{DF}+\text{BF}}$

so when $\frac{\text{BDF}}{\text{BD}+\text{DF}+\text{BF}} \approx B > 295$, we are compute-bound for the NVIDIA H100 SXM